

MATH 189 - Data Analysis

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1 Statistical Learning

1.1 Terminology

"Supervised Learning" means we have both predictors and responses.

"Predictors/input variables/features" are arranged in a vector.

$$X = (X_1, X_2, \dots, X_n)$$

"Output variables/responses" is a scalar where $y \in \mathbb{R}$

We think of data as coming from a model.

$$y = f(x) + \epsilon$$

$\begin{cases} f(x) \text{ is fixed but unknown function} \\ \epsilon \text{ is the random error term with mean zero, independent of } X \end{cases}$

Question: Why estimate f ?

Prediction: Given X , we want to predict y with $\hat{y} = \hat{f}(x)$

Mean square error is

$$\begin{aligned} \mathbb{E}(y - \hat{y})^2 &= \mathbb{E}(f(x) + \epsilon - \hat{f}(x))^2 \\ &= \mathbb{E}(f(x) - \hat{f}(x))^2 + 2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) + \mathbb{E}(\epsilon^2) \end{aligned}$$

Examine the middle term in detail

$$2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) = \mathbb{E}(f(x) - \hat{f}(x)) \cdot \mathbb{E}(\epsilon) = 0$$

We know that

$$\mathbb{E}(y - \hat{y})^2 = (f(x) - \hat{f}(x))^2 + \mathbb{E}(\epsilon^2)$$

Where the left term is reducible error and the right term is irreducible error.

"Inference" - Sometimes we want to understand the association between y and $X_1, \dots, X_p$

1. Which predictors are associated with response?
2. Is the association positive or negative?
3. Is the relationship approximately linear?

1.2 Accuracy

Training data

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

where $X_i = (X_{i1}, X_{i2}, \dots, X_{ip})$

n = # data points (e.g. $n = 30$)

p = # predictors (e.g. $p = 3$)

Mean Square Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

Defined on training data

$$\text{Test MSE} = \text{Ave } (y_0 - \hat{f}(x_0))^2$$

Defined on previously unseen data point (x_0, y_0)

Linear / affine model

$$f(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

”Fit/train the model” usually means run an optimization on your computer that uses training data and outputs the parameters $\beta_0, \beta_1, \dots, \beta_p$ that make the MSE and hopefully the test MSE small.

Classification

Data set $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ where $X_i = (x_{i1}, \dots, x_{ip})$ and y_i is qualitative.

Error rate is the fraction of incorrect classifications.

$$\frac{1}{n} \sum_{i=1}^n \mathbb{I}\{y_i \neq \hat{y}_i\}$$

$$\mathbb{I}\{y_i \neq \hat{y}_i\} = \begin{cases} 1, & y_i \neq \hat{y}_i \\ 0, & y_i = \hat{y}_i \end{cases}$$

Test error rate

$$\text{Ave}(\mathbb{I}\{y_0 \neq \hat{y}_0\})$$

Where (x_0, y_0) is a previously unseen test data point.

K-nearest neighbors (KNN)

The KNN classifier is defined by letting η_0 = set of indices $i \in \{1, \dots, n\}$ for K nearest neighbors of X_0 among the input data.

$\hat{y}_0 = j$ where j maximizes

$$\sum_{i \in \eta_0} \mathbb{I}\{y_i = j\}$$

which is the number of nearest neighbors (out of K) assigned label j .

2 Linear Regression

2.1 Simple Linear Regression

Underlying statistical model: $y = \beta_0 + \beta_1 X + \epsilon$ where y is the quantitative response and a single predictor X which is why it is called simple.

Prediction model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$

$\hat{\beta}_0, \hat{\beta}_1$ are estimates based on training data, where hat means estimated quantity.

Estimating coefficients

The residual sum of squares (RSS)

$$\begin{aligned} \text{RSS} &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \end{aligned}$$

Choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize RSS.

$$\begin{aligned}\frac{\partial \text{RSS}}{\partial \hat{\beta}_0} &= \frac{\partial}{\partial \hat{\beta}_0} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \\ &= 2 \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\ \implies \text{RSS is } &\begin{cases} \text{decreasing for really small } \hat{\beta}_0 \\ \text{increasing for really big } \hat{\beta}_0 \end{cases}\end{aligned}$$

RSS is optimal (in $\hat{\beta}_0$) when

$$\begin{aligned}0 &= \frac{1}{n} \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\ &= \hat{\beta}_0 + \hat{\beta}_1 \left(\frac{1}{n} \sum_{i=1}^n X_i \right) - \left(\frac{1}{n} \sum_{i=1}^n Y_i \right) \\ &= \hat{\beta}_0 + \hat{\beta}_1 \bar{x} - \bar{y} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}\end{aligned}$$

Where $\begin{cases} \bar{x} = \frac{1}{n} \sum_{i=1}^n X_i \text{ is the mean predictor} \\ \bar{y} = \frac{1}{n} \sum_{i=1}^n Y_i \text{ is the mean response} \end{cases}$

Covariance

$$\begin{aligned}\text{Cov}(X_i, Y_i)_{i=1}^n &= \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y} \\ &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})(Y_i - \bar{y})\end{aligned}$$

Variance

$$\begin{aligned}\text{Var}(X_i)_{i=1}^n &= \text{Cov}(X_i, X_i)_{i=1}^n \\ &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})^2\end{aligned}$$

From this we can get our optimal $\hat{\beta}_1$

$$\begin{aligned}\hat{\beta}_1 &= \frac{\text{Cov}(X_i, Y_i)_{i=1}^n}{\text{Var}(X_i)_{i=1}^n} \\ &= \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})(Y_i - \bar{y})}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})^2}\end{aligned}$$

2.2 Goodness of Fit & Correlation

Assume an underlying statistical model

$$y = \beta_0 + \beta_1 X + \epsilon$$

We use training data $(X_i, Y_i)_{i=1}^n$ to generate prediction model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$
The formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize RSS:

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \begin{cases} \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{X} \\ \hat{\beta}_1 = \frac{\text{Cov}(X_i, Y_i)_{i=1}^n}{\text{Var}(X_i)_{i=1}^n} \end{cases}$$

Goodness of fit

$$\begin{aligned}\text{RSS} &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \\ &= \sum_{i=1}^n \left((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}) \right)^2 \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 - 2\hat{\beta}_1 \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{\left[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \right]^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\end{aligned}$$

RSS scales with the number of data points n and also scales with y^2 units.
→ We can fix the scaling and get out a goodness-of-fit score (R^2) between 0 and 1.

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \qquad \text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$$

So for simple linear regression,

$$\begin{aligned}
 R^2 &= 1 - \frac{\text{RSS}}{\text{TSS}} \\
 &= \frac{\text{TSS} - \text{RSS}}{\text{TSS}} \\
 &= \frac{\left[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \right]^2}{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2} \\
 R &= \frac{\text{Cov}((x_i), (y_i))}{\text{Var}(x_i)^{1/2} \text{Var}(y_i)^{1/2}}
 \end{aligned}$$

Long story short R^2 is the fraction of variance explained by the model.

ex $\hat{\text{sales}} = 7.0 + 0.048 \cdot \text{TV}$
 $R^2 = 0.61 = 61\%$ $r \approx 0.78$ pretty strong positive correlation

2.3 Regression Table

ex

	Coefficient	Std. error	t-statistic	p-value
Intercept (β_0)	7.0	0.46	15	<0.0001
TV (β_1)	0.048	0.0027	18	<0.0001

Std error is the error bar on linear coefficients.

We know that $\begin{cases} \beta_0 \approx \hat{\beta}_0 \pm \text{SE}(\hat{\beta}_0) \\ \beta_1 \approx \hat{\beta}_1 \pm \text{SE}(\hat{\beta}_1) \end{cases}$

Assume the underlying statistical model is true!

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where ϵ_i are independent $\eta(0, \sigma^2)$ random variables

If we generate N data points (x_i, y_i) from the model, then

$$\begin{cases} \text{Var}(\hat{\beta}_0)^{1/2} \approx \text{Std. error}(\hat{\beta}_0) \\ \text{Var}(\hat{\beta}_1)^{1/2} \approx \text{Std. error}(\hat{\beta}_1) \\ \mathbb{E}(\hat{\beta}_0) = \beta_0 \\ \mathbb{E}(\hat{\beta}_1) = \beta_1 \end{cases}$$

2.4 Multiple Linear Regression

The underlying statistical model that we assume is true.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

where y is response, β_0 is the intercept, $\beta_1 x_1, \dots, \beta_p x_p$ are predictors, and ϵ is the mean zero noise.

$$\boxed{ex} \quad \text{sales} = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \text{Newspaper} + \epsilon$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

We want to find the best $\hat{\beta}$ to minimize RSS, where $\hat{\beta}$ are estimated quantities.

$$\begin{aligned} RSS &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip})^2 \\ &= \left\| \begin{bmatrix} y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_{11} - \dots - \hat{\beta}_p x_{1p} \\ \vdots \\ y_n - \hat{\beta}_0 - \hat{\beta}_1 x_{n1} - \dots - \hat{\beta}_p x_{np} \end{bmatrix} \right\|^2 \\ &= \|Y - X\hat{\beta}\|^2 \end{aligned}$$

$$\text{where } y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{1p} \\ & & \ddots & \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}$$

The best $\hat{\beta}$ to minimize Euclidean norm² for $\|Y - X\hat{\beta}\|^2$ is $\hat{\beta} = X^+ Y = (X^T X)^{-1} X^T Y$

	Coefficient	Std. error	t-statistic	p-value
Intercept	$\hat{\beta}_0$ 2.939	0.3119	9.42	<0.0001
TV	$\hat{\beta}_1$ 0.046	0.0014	32.81	<0.0001
Radio	$\hat{\beta}_2$ 0.189	0.0086	32.89	<0.0001
Newspaper	$\hat{\beta}_3$ -0.001	0.0059	-0.18	0.8599

Note: the standard error is NOT $Var(\hat{\beta}_i)^2$ but rather $\hat{Var}(\hat{\beta}_i)^2$

Hypothesis testing

State a hypothesis, test it with data (testing the likelihood such data could possibly be generated if the hypothesis were true).

H_0 - the null hypothesis: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$ for some coefficients $\beta_0, \beta_2, \beta_3$ with $\beta_1 = 0$ with $\epsilon \sim N(0, \sigma^2)$ for some variance $\sigma^2 \geq 0$. Also, observations are independent.

H_1 - the alternative hypothesis: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$ where everything above holds except $\beta_1 \neq 0$

Assume H_0 . Then

$$\frac{\hat{\beta}_1}{\hat{SE}(\hat{\beta}_1)} = t \sim T_{n-p-1}$$

3 Logistic Regression & Generative Models

3.1 Logistic Regression

We want to predict whether an event happens ($y = 1$) or not ($y = 0$), we really want to model the probability $p \in (0, 1)$ of occurrence $\implies$ NOT LINEAR REGRESSION.

Before

$$y = \beta_0 + \beta_1 X_\epsilon$$

Now

$$p = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$

$$1 - p = \frac{1}{1 + \exp(\beta_0 + \beta_1 X)}$$

$$\frac{p}{1 - p} = \exp(\beta_0 + \beta_1 X)$$

Where the term $\frac{p}{1-p}$ is called "odds".

Note: $\log(\text{odds}) = \beta_0 + \beta_1 X$

ex Defaulting on bank loans

p = probability of defaulting

X = balance

$$y_i = \begin{cases} 0 & \text{default doesn't happen} \\ 1 & \text{default does happen} \end{cases}$$

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \cdot X$$

How do we use data/observations $(x_i, y_i)_{1 \leq i \leq n}$ to best estimate β_0, β_1 .

$$\begin{cases} \text{Linear regression} \implies \text{minimize RSS} \\ \text{Logistic regression} \implies \text{use max likelihood, do it on a computer} \\ \text{Generative models} \implies \text{simple formulas for } \hat{\Pi}_k, \hat{\mu}_k, \hat{\Sigma}_k \end{cases}$$

We choose estimates $\hat{\beta}_0, \hat{\beta}_1$ to maximize likelihood of observations $(x_i, y_i)_{1 \leq i \leq n}$ if (a) the observations were independent and (b) they were generated exactly from the statistical model.

$$\begin{aligned} \mathbb{P}((x_i, y_i)_{1 \leq i \leq n}) &= \prod_{i=1}^n \mathbb{P}(x_i, y_i) \text{ where } y_i \in \{0, 1\} \\ &= \prod_{y_i=1} \hat{p}(x_i) \prod_{y_i=0} (1 - \hat{p}(x_i)) \\ &= \prod_{i:y_i=1} \frac{\exp(\beta_0 + \beta_1 X_i)}{1 + \exp(\beta_0 + \beta_1 X_i)} \prod_{i:y_i=0} \frac{1}{1 + \exp(\beta_0 + \beta_1 X_i)} \end{aligned}$$

Know that $\hat{\beta}_0, \hat{\beta}_1$ are chosen to maximize this probability/likelihood above.

ex $P_{\text{defaulting}}$

	Coefficient	Std. error	z-statistic	p-value
Intercept	-10.6513	0.3612	-29.5	<0.0001
Balance	.0055	0.0002	24.9	<0.0001

We want $\hat{\beta}_0 + \hat{\beta}_1 X = 0$

$$-10.6513 + 0.0055X = 0$$

$$X = \frac{10.6513}{0.0055} \\ = 1936.6$$

Assume

- a. The model is true
- b. Observations $(x_i, y_i)_{1 \leq i \leq n}$ are independent draws from the model
- c. Model is true with $\beta_0 = 0$ (null hypothesis)

3.2 Generative Models for Classification

Predictors

X or $X_1, X_2, \dots, X_n$

Response

$y \in \{0, 1\}$ or $y \in \{1, 2, \dots, k\}$

Review

Simple logistic regression

$$P = \text{prob}\{y = 1\} = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$

Note: This doesn't make any assumptions about how predictors are distributed.

For each $k = 1, 2, \dots, K$:

Our statistical model ('mixture of Gaussians') says $y = k$ with probability π_k ($\sum_{k=1}^K \pi_k = 1$)

Given $y = k$, $X \sim N(\mu_k, \Sigma_k)$ where Σ is the covariance matrix symbol. Here is the general vector-valued format:

$$X = (X_1, \dots, X_p) \quad \Sigma_k = \begin{bmatrix} & \end{bmatrix} \text{ pxp matrix}$$

Where μ_k has Kp parameters and Σ_k has $K\frac{p(p+1)}{2}$ parameters

When $p = 1$, we can write

$$X \sim N(\mu_k, \sigma_k^2)$$

In approaches like Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and Naive Bayes (NB) use simple estimators for Π_k, μ_k, Σ_k that we can calculate fast.

ex Linear discriminant analysis (LDA) make assumptions of common covariance structure $\Sigma = \Sigma_1 = \Sigma_2 = \dots = \Sigma_k$ which reduces a lot of complexity. Given observations $(X_i, y_i)_{1 \leq i \leq n}$, we can estimate for $k = 1, \dots, K$

$$\begin{aligned}\hat{\Sigma} &= \frac{n_k}{n}, \text{ where} \\ n &= \#\{i : y_i = k\} \\ &= \text{number of observations in class } k \\ \hat{\mu}_k &= \frac{1}{n_k} \sum_{i: y_i = k} X_i\end{aligned}$$

Last estimate common covariance by

$$\begin{aligned}\hat{\Sigma} &= \frac{1}{n - K} \sum_{k=1}^K \sum_{i: y_i = k} (X_i - \mu_k)(X_i - \mu_k)^T \\ \mathbb{E}\hat{\Sigma} &= \Sigma \text{ if the model is true}\end{aligned}$$

ex Quadratic discriminant analysis is pretty much the same except DIFFERENT covariances $\Sigma_1, \dots, \Sigma_k$ for each class.

$$\begin{aligned}\implies \hat{\Pi}_k &= \frac{n_k}{n}, \hat{\mu}_k = \frac{1}{n_k} \sum_{i: y_i = k} X_i \\ \implies \hat{\Sigma}_k &= \frac{1}{n_k - 1} \sum_{i: y_i = k} (X_i - \mu_k)(X_i - \mu_k)^T \\ \implies &K\frac{p(p+1)}{2} \text{ covariance free parameters.}\end{aligned}$$

ex Naive Bayes is pretty much the same except DIFFERENT covariances $\Sigma_1, \dots, \Sigma_k$ that are also diagonal matrices.

$\implies Kp$ covariance free parameters with the same estimators as QDA.

$$\text{diag} \left(\sum_k^n \right) = \frac{1}{n_k - 1} \sum_{i: y_i = k} (X_i - \mu_k) \odot (X_i - \mu_k)$$

Suppose you have the statistical model with probability Π_k , we set $y = k$ and draw $X \sim N(\mu_k, \Sigma_k)$. Suppose we have estimated the parameters well - see a new predictor vector $X = (X_1, \dots, X_p)$ - How do you determine class probabilities $P\{y = k|X\}$ and make predictions?

Prediction part is easy - just predict based on the highest probability class:
 $\hat{y} = k$ for $k = \operatorname{argmax}_{1 \leq j \leq K} P\{y = j|X\}$

So what is $P\{y = k|X\}$? Bayes' rule tells us

$$\begin{aligned} P\{y = k|X\} &= \frac{P\{x|y = k\} P\{y = k\}}{\sum_{j=1}^K P\{X|y = j\} P\{y = j\}} \\ &= \frac{f_k(X) \Pi_k}{\sum_{j=1}^K f_j(X) \Pi_j} \end{aligned}$$

where Gaussian density $\Pi_j(X) = \frac{1}{(2\pi)^{p/2}(\det \Sigma)^{1/2}} \exp(-\frac{1}{2}(X - \mu_j)\Sigma_j(X - \mu_j))$

3.3 Discriminant Analysis In Detail

Let's say we have a generative statistical model with probability $\frac{1}{2}, y = 1$.

Otherwise, $y = 2$.

Given $y = 1, X \sim N(\mu_1, \sigma^2)$

Given $y = 2, X \sim N(\mu_2, \sigma^2)$

We think about probability of class membership using Bayes' rule.

$$P\{Y = k|X\}$$

Where Y is response, k is class, either 1 or 2, and X is the predictor.

$$\begin{aligned} &\frac{P\{X, Y = k\}}{P\{X\}} \\ &= \frac{P\{X, y = k\}}{\sum_{j=1}^2 P\{X, y = j\}} \\ &= \frac{P\{y = k\}P\{X|y = k\}}{\sum_{j=1}^2 P\{y = j\}P\{X|y = j\}} \end{aligned}$$

Plugging in the specific model for class 1,

$$\begin{aligned}
 P\{y = 1|X\} &= \frac{\frac{1}{2}f_1(X)}{\frac{1}{2}f_1(X) + \frac{1}{2}f_2(X)} \\
 \text{where } f_1(x) &= P\{X|y = 1\} \\
 &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(X - \mu_1)^2\right) \\
 f_2(x) &= P\{X|y = 2\} \\
 &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(X - \mu_2)^2\right)
 \end{aligned}$$

Usually our task in this class is to predict y as well as possible $\hat{y} = 1$ or $\hat{y} = 2$. The best we can do, the only normal strategy, is to predict the higher probability count.

$$\hat{y} = 1 \text{ if } P\{y = 1|X\} > P\{y = 2|X\}$$

Which happens if and only if

$$\begin{aligned}
 \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(X - \mu_1)^2\right) &> \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(X - \mu_2)^2\right) \\
 \exp\left(-\frac{1}{2\sigma^2}(X - \mu_1)^2\right) &> \exp\left(-\frac{1}{2\sigma^2}(X - \mu_2)^2\right) \\
 -\frac{1}{2\sigma^2}(X - \mu_1)^2 &> -\frac{1}{2\sigma^2}(X - \mu_2)^2 \\
 (X - \mu_2)^2 &> (X - \mu_1)^2
 \end{aligned}$$

So we make the prediction, $\begin{cases} \hat{y} = 1 & \text{if } |X - \mu_1| < |X - \mu_2| \\ \hat{y} = 2 & \text{if } |X - \mu_2| < |X - \mu_1| \end{cases}$

We want to move from 1 predictor X to two predictors X_1 and X_2

Before

$P\{y = 1\} = P\{y = 2\} = \frac{1}{2}$
 Given $y = 1$, $X \sim N(\mu_1, \sigma^2)$
 Given $y = 2$, $X \sim N(\mu_2, \sigma^2)$

After

$$P\{y = 1\} = P\{y = 2\} = \frac{1}{2}$$

Given $y = 1$, $X \sim N(\mu_1, \sigma^2)$

Given $y = 2$, $X \sim N(\mu_2, \sigma^2)$

Where $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ and $\vec{\mu}_1 = \begin{bmatrix} \mu_{11} \\ \mu_{12} \end{bmatrix}$ and $\vec{\mu}_2 = \begin{bmatrix} \mu_{21} \\ \mu_{22} \end{bmatrix}$

Making it bold to turn scalars into vectors works except with variance.

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

4 Model Selection

4.1 Bootstrap

Quantitative response - "train" different multiple linear regression models with different predictors.

Qualitative response - LDA / QDA / NB multiple logistic regression models with different predictors.

To get the best model, consider:

1. Set aside validation data set

$$\begin{array}{ccc} & (X_i, y_i)_{1 \leq i \leq n} & \\ & \swarrow \quad \searrow & \\ (X_i, y_i)_{i \in T} & & (X_i, y_i)_{i \in T^C} \end{array}$$

Where T is a set of training data point indices, e.g. $T = \{1, \dots, 100\}$ and T^C is the "test/validation" data points, e.g. $T^C = \{101, \dots, 110\}$. $\hat{y}_i$ is the prediction for input X_i often using training data $(X_i, y_i)_{i \in T}$ to train the model. We can measure $\text{MSE}_{\text{test}} =$
 $\Rightarrow$ Choose the model with the best (lowest) MSE_{test}

2. Leave-one-out cross-validation (LOOCV)
 Idea: use all possible train-test splits.

$$(X_j, y_j)_{j \neq i}$$

3. K-fold cross validation

Split your data according to k index sets $\{1, \dots, n\} = S_1 \cup S_2 \cup \dots \cup S_k$

4. Bootstrap

$\boxed{ex}$ you invest $\begin{cases} \alpha & \text{fraction of your money in asset with return value } X \\ 1 - \alpha & \text{fraction in asset with return value } y \end{cases}$

You want to minimize variance of your return.

Choose α to minimize

$$\text{Var}[\alpha X + (1 - \alpha)y] = \text{Cov}[\alpha X + (1 - \alpha)y, \alpha X + (1 - \alpha)y]$$

To optimize, set α derivative equal to zero.

$$\begin{aligned} \alpha &= \frac{\text{Cov}[y-x, y]}{\text{Var}[y-x]} \\ &= \frac{\sigma_y^2 - \sigma_{xy}}{\sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}} \end{aligned}$$

In practice, we use data $(x_i, y_i)_{1 \leq i \leq n}$ to estimate

$$\begin{aligned} \hat{\alpha} &= \frac{\hat{\sigma}_y^2 - \hat{\sigma}_{xy}}{\hat{\sigma}_x^2 + \hat{\sigma}_y^2 - 2\hat{\sigma}_{xy}} \\ \text{where } \begin{cases} \hat{\sigma}_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \\ \hat{\sigma}_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 \\ \hat{\sigma}_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \end{cases} \end{aligned}$$

Q: What is a good error bar for $\hat{\alpha}$?

Q: what is $\text{SE}[\hat{\alpha}] = \text{Var}[\hat{\alpha}]^{\frac{1}{2}}$?

Bootstrap answers this.

Bootstrap with $B(\approx 10^2 - 10^9)$ samples

For $j = 1, \dots, B$:

Make fake data set $(\tilde{x}_i, \tilde{y}_i)_{1 \leq i \leq n}$ that comes from real observations $(x_i, y_i)_{1 \leq i \leq n}$ by uniformly sub-sampling with replacement.

4.2 Linear Model Selection

Goal: Choose best linear regression model based on predictor selection and modification of predictors.

Question: Given a data set with p predictors and a quantitative response variable, which subset of predictors leads to the best multiple linear regression model?

1. Best subset selection

- Try all 2^p models and evaluate the MSE on a validation data set.
- Choose the best one.

2. Forward selection

- Start with an empty set of variables.
- Try adding each of the remaining variables and calculate the MSE on a validation set.
- Choose the best variable and add it to the model if it improves the MSE.
- Repeat until convergence.

3. Backward selection

- Start with all p variables.
- Try kicking out each of the variables and calculate MSE on a validation set.
- If it improves MSE, make the best alteration to the model.
- Repeat until convergence.